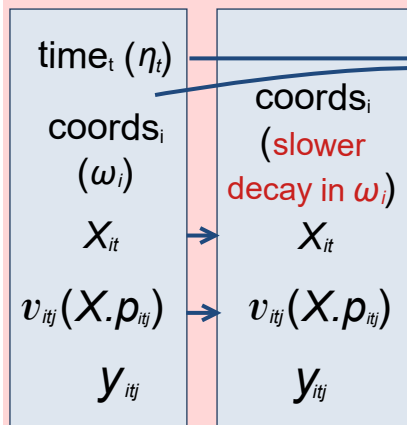


STUDY 1

Baseline with
Bernoulli design
Stronger
autocorrelation

1-0

1-1



NAs

NAs

J=2

$$G_{itj} \sim \mathcal{B}(p)$$

STUDY 2

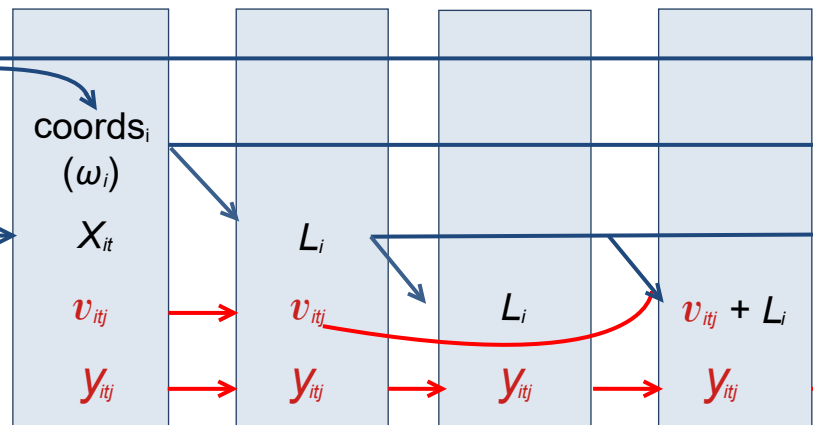
Baseline with
Poisson design
Latitude as
occupancy covariate
Overlap of covariates
Partial overlap of
covariates

2-0

2-1

2-2

2-3



NAs

NAs

NAs

NAs

J=10

$$D_{it} \sim \text{Poisson}(\lambda = 1.1)$$

$$G_{itj} = \begin{cases} 1 & \text{if } j \in \mathbf{r}_j \\ 0 & \text{otherwise} \end{cases}$$

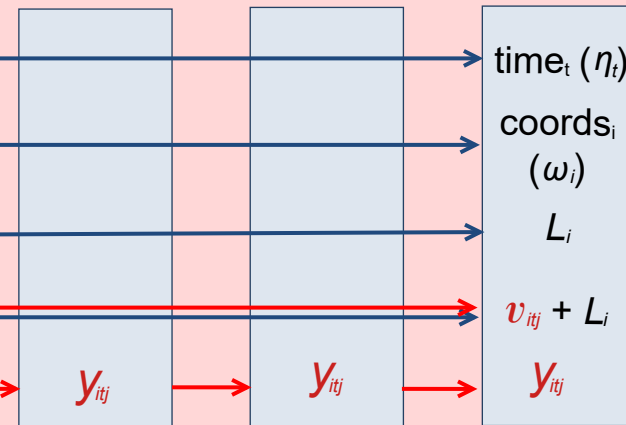
STUDY 3

Phenology effect
Observation spot
(25%)
Observation spot
(Data amount
reestablished)

3-1

3-2

3-3



NAs

NAs

NAs

$$D_{it} \sim \text{Poisson}(\lambda = 1.1 \times (I/I^*))$$

Legend

— simTOcc
— custom
— NAs